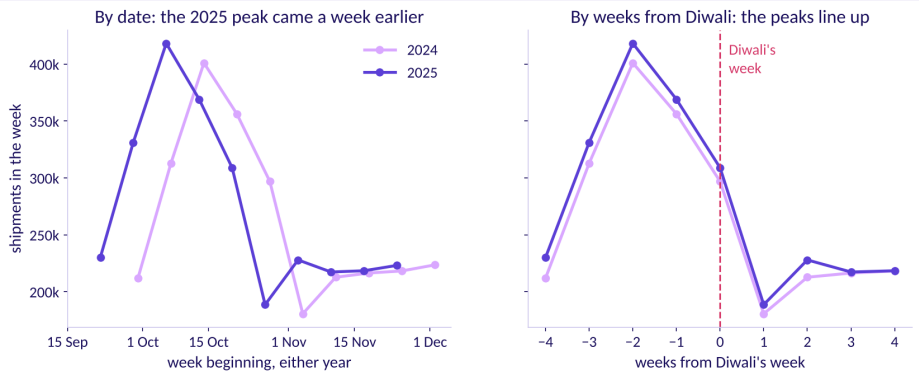
CHEAT SHEET

How much is coming, and how wrong could the forecast be?

Flipkart, the Lucknow fulfilment centre, Week 4 Friday. The planner books packer shifts for the eight weeks beginning 12 October to 30 November 2026; a vendor promises 22 percent above last year and finance 4 percent. Illustrative case: the company is real, the situation and the data are synthetic, built for this class.

C2_W04_D05_beat_the_bot_STUDENT
Verified 12 Oct 2026
7 panels, 1 diagram

1 Where does the peak fall, and which rule wins on hidden weeks?



Rule	MAE on 13 hidden weeks	Misses priced
Festival-aligned	6,615	Rs 13.5 lakh
Finance's rule	6,615	Rs 13.5 lakh
House bot	9,831	Rs 30.5 lakh
Drift	14,842	Rs 28.0 lakh
Naive	14,923	Rs 28.8 lakh
Vendor's curve	16,400	Rs 11.9 lakh

Crux. Both years peak two weeks before Diwali's week, and only a copy that counts weeks from Diwali, scored on weeks it never saw, earns the plan.

2 What do twelve weeks say, counted by hand?

Weeks from Diwali's week	-3	-2	-1	Diwali	+1	+2
2024	312,900	401,000	356,200	296,900	180,600	212,700
2025	331,200	418,200	369,000	308,900	188,600	227,900

For Diwali 2025's week, 308,900 shipped: naive said 369,000, 60,100 too high; the house bot 356,200, 47,300 too high; festival-aligned 296,900 x 1.04 = 308,776, 124 too low.

Crux. A copy of the calendar lands a moving festival in the wrong week; a copy by distance from Diwali lands it.

3 What does each rule copy?

Rule	The forecast	In words
Naive	y[T]	Repeat the last week seen
House bot	y[t - 52]	The same week last year
Drift	y[T] + h x weekly change	The line from week one, extended
Finance's rule	1.04 x y[t - 52]	The house bot plus 4 percent
Festival-aligned	1.04 x the week at the same distance from last Diwali	Moves with the festival

y[t] is week t's shipments, T the last week a rule may see, h the weeks ahead. Festival-aligned matches distance from Diwali from five weeks before Diwali's week to four after.

Crux. Every rule copies something from the past, so say what it copies before trusting what it says.

4 How is a forecast scored?

- Hide the latest weeks, at least as many as the forecast must reach ahead, ten here: the test hides 13, 29 June to 21 September 2026, forecast from 22 June.
- MAE is the average of the misses with their signs dropped, in shipments.
- Price each miss by its direction: Rs 5.42 for a spare shipment, Rs 25 for a late one.
- The vendor's curve missed its 91 fitted weeks by at most 1.4 percent, and the hidden weeks by 16,400 a week.

Crux. Beat the bot on hidden weeks: in shipments, then rupees, against a copy of last year, before any model earns the plan.

5 Where does each forecast put the festival in 2026?

Forecast	Busiest week	Against 13 Oct to 1 Dec 2025	Against 29 Sep to 17 Nov 2025
House bot	5 Oct	the same	13.0 percent below
Finance's rule	5 Oct	4.0 percent above	9.5 percent below
Vendor's curve	5 Oct	22.0 percent above	6.1 percent above
Festival-aligned	19 Oct	19.5 percent above	4.0 percent above

On 2025's festival, forecast from 22 September 2025, the house bot missed by 58,250 a week and festival-aligned by 4,034.

Crux. Copy by distance from Diwali: a calendar copy moved 2025's peak a week late and moves 2026's two weeks early.

6 Where should the plan sit, and how many shifts?

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share of weeks covered = late / (late + spare) = 25 / (25 + 5.42) = 0.82
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- Why: over 100 weeks, if demand beats the plan in 18, one more shipment of plan saves 18 x Rs 25 = Rs 450 and wastes 82 x Rs 5.42 = Rs 444.
- Sorted, the 13 hidden weeks put the 0.82 point between the 10th (2.1 percent above the forecast) and the 11th (2.7): 2.6 percent, which covers 10 of 13, the 11th just above it.
- Shifts are festival-aligned x 1.026 over 120, rounded up: 20,275 in eight weeks, 3,719 in the week of 19 October.
- The error bar is the forecast's average miss, 6,615 shipments a week: about Rs 35,800 when it runs that far high, Rs 1.65 lakh when that far low.

Crux. Cover 0.82 of weeks: a late shipment costs 4.6 times a spare one, so the plan sits 2.6 percent above the forecast.

7 What does the interviewer listen for?

"I hold out the latest weeks, at least as many as the forecast must reach ahead, and score the misses in shipments and rupees against the seasonal naive: festival-aligned missed by 6,615 a week, the seasonal naive by 9,831." An AI company sizes launch capacity in tokens a second from the ground up, checks it against earlier launches by days from launch with the weekday kept, and books above the forecast to a latency target.

Crux. Name the hidden weeks, the unit, the copy to beat and the direction that costs more.

THE WORDS ON THIS SHEET

House bot A copy of the same week last year

Festival-aligned A copy at the same distance from last year's Diwali, plus 4 percent

Hold-out Hidden weeks, kept out of every forecast and used to score it

Origin and horizon The last week a rule may see, and the farthest week a forecast must reach

Back-test A rule rerun from an earlier date

MAE and MAPE The average miss in shipments, and as a percent of what shipped

Error bar How wrong the forecast could be: its average miss on unseen weeks

Plan share The share of weeks the plan should cover